

A rock and a hard place: Setting up OpenGL

OpenGL is the primary requirement for playing 3D games on Linux, and usually comes pre-installed. If this isn't the case, you have the task of installing it. OpenGL comes in various flavours. One of the popular ones is the freely available Mesa3D variant, available at www.mesa3d.org, or at <http://mesa3d.sourceforge.net>. You need to download the tarball package of the latest version, and install it. There are two parts to download—MesaLib and MesaDemos—of which only MesaLib is required to play OpenGL games: MesaDemos only contains all the demos that come along with the Mesa library.

Untar it, and then run the following command:

```
./configure --prefix=/usr
```

For Redhat 8.0, 9.0 and Mandrake 9.1, the following command should be used:

```
export LDFLAGS="-lstdc++"
```

```
./configure --prefix=/usr
```

After this, use the following commands:

```
make
```

```
make check
```

```
make install
```

These steps will install the Mesa libraries to their required destinations. Do this carefully, as even small mistakes could lead to major errors during installation.

Mountains and molehills: The games

There are many games available for Linux; here are some of the more popular ones, and the steps required to set them up.

Quake III

It's been quite a while since *Quake* was ported to Linux. It is also one of the few games with very few problems and extensive documentation. There's a very comprehensive 'How To' document on the Linux Documentation project Web page. Installation is simple—just download the rpm package,



Quake was one of the first popular games to be ported to Linux

The X Factor

Though Xbox is the gaming console from the Microsoft world, diehard Linux junkies have found a way to hack into it and install Linux. There is a very active project called the Xbox Linux project, which exists with the sole purpose of making a Linux distribution for the Xbox. So far the results have been pretty impressive.

Why would anybody want to install Linux on an Xbox, you ask? Put simply, an Xbox costs about \$179, and has a Pentium III 733 MHz processor with a signifi-

cantly superior graphics sub-system and loads of memory. It's the ideal solution to cheap Linux boxes! As for all those pretty games, Linux geeks have figured out a way to dual boot the Xbox, so you now have a cheap, stable and fast Linux box that can morph into a gaming monster at will. Another interesting point that diehard Linux fans make is the Xfactor: anything called 'X' is related to Linux or Unix, and so the Xbox should be renamed LinuXbox!

and use the '**rpm -ivh**' command to install it. This will create a **/usr/games/quake3** directory, and place all the base files there. All the commands will also be installed in the proper locations. Just type '**quake3**' at the command prompt to start the game.

TORCS

TORCS is one of the best OpenGL-based 3D racing simulation on Linux. Installing TORCS is slightly tricky. First download the TORCS-1.2.1-src.tgz package from www.torcs.org, or torcs.sourceforge.net. If you need opponents (robots) to play with, download the three robot packages (TORCS-1.2.1-src-robots-base.tgz, TORCS-1.2.1-src-robots-K1999.tgz and TORCS-1.2.1-src-robots-berniw.tgz). Now, untar all



TORCS is a pure Linux game, and one of the best racing games today

these packages, then '**cd**' to the torcs-1.2.1 directory, type '**./configure**', then '**make**', and finally '**make install**'. This will compile the game and install it in **/usr/games/torcs**. Next, download all the data packages—TORCS-1.2.1-data.tgz, TORCS-1.2.1-data-tracks-base.tgz, TORCS-1.2.1-data-cars-extra.tgz, and TORCS-1.2.1-data-cars-Patwo-Design.tgz. You only need the first data package to play the game, but the other packages make the game more interesting. Copy the data packages (the .tgz files) to the **/usr/games/torcs** directory,

and untar them. After this the game is ready, and you can start playing it using the '**torcs**' command.

Unreal Tournament

It's been a long time since *Unreal*, too, was ported to Linux. The older *Unreal* demos are available for download from www.unrealtournament.com. The download file, utde-



Once you install UT2003, just pop in your UT2003 CD to play the game

mo-linux-x86-348.tar.gz, is not source code, so just untar it into the **/usr/games** directory—no need to compile.

For UT2003, just download the installer script from www.ut2003.com and run the script with the game CD in the CD-ROM drive. You'll have to mount the CD and run the 'give execute rights' for the CD using '**chmod**', to play the game.

Nuclear chess

Nuclear chess is a chess variant that is great fun, whether you can play chess or not. In this game, when one piece captures another, both pieces as well as the neighbouring pieces are lost, making each game short and loads of fun.

Installing nuclear chess is quite simple—just download **nuclearchess 0.9.2.tar.gz** from www.linux-games.com, untar it, and compile the source code.